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Do Tourists Tip More Than Local Consumers? Evidence from Taxi Rides in New York City

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Abstract

Given the resurgence of cities as consumer centers and the importance of amenities, we revisit the differences in tipping in taxis between tourists and locals in New York City. Taxi service is an endogenous service; however, taxis also contribute to the demand and provision of other amenities. We compare locals and tourists who are theatergoers to control for education and income, as these factors are likely to affect tipping behavior. Using data from the New York City Taxi and Limousine Commission on yellow taxis, we identify tourists as those trips leaving from or going to a hotel and theatergoers as trips where the drop-off or pickup is near Broadway within thirty minutes of the beginning or end of a show. We find that tourists and theatergoers tip more than locals and nontheatergoers, and tourists who are theatergoers tip even more. These differences between tourists and locals may affect the allocation of taxis throughout the city and hence the provision of other amenities.

Keywords

taxi, tourists, tipping difference, amenities, spatial allocation

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The resurgence of cities is well established in the economics literature (Clark et al. 2002; Glaeser and Gottlieb 2006). The emergence of urban spaces, which in the past was mainly driven by productive forces, such as the clustering of firms to enjoy agglomeration economics, is now being driven by a different engine: consumption and amenities. Glaeser, Kolko, and Saiz (2001); Glaeser and Gottlieb (2006); and Albouy (2016), building on the Rosen (1979) and Roback (1982) framework, provide evidence of the importance of amenities—including both services and consumption goods—to city growth. For instance, Glaeser, Kolko, and Saiz (2001) state that cities with more restaurants and live performance theaters per capita experienced larger growth rates during the 1980s and 1990s in both the United States and France. Brueckner, Thisse, and Zenou (1999) expand the monocentric city model developed by Alonso (1964), Mills (1967), and Muth (1969) to include what the authors call exogenous and endogenous amenities. The former includes things such as natural or historical amenities, which are generally fixed to a location. The latter is the so-called modern amenities, such as transportation systems, restaurants, and cultural amenities, are dependent on local demand.

Regarding the use of cultural amenities, Alderighi and Lorenzini (2012) follow Becker's rational addiction model and find that to gain some cultural capital, consumers in general are willing to sacrifice some current utility to consume these goods. Castiglione and Infante (2016) find evidence that supports the rational addiction model for theatergoers in Italy. However, the demand for modern amenities, including theater and other culture amenities, can be studied by differentiating the patterns of consumption from locals and tourists, as these groups may have different demands for these goods and services. For instance, Bonet (2003) supports the existence of such demand for cultural amenities from tourists, and Gapinski (1988) showed that tourists are around 65 percent of attendees of performing arts shows in London. Zieba (2015), in turn, stressed that foreigners and non-German tourists have a positive impact on demand in Austrian theaters, while domestic tourists did not have an impact. Also, Nicolau (2010) argues that tourists are price insensitive with respect to cultural interests.

In this article, we look at the differences in tipping between tourists and locals in the taxi industry in New York City (NYC). Taxi service is an endogenous amenity, thus the different demands and tipping behavior of the two groups may affect the allocation of cabs throughout a city. Tourists are unlikely to have repeated interactions with service workers in the area. While statistically speaking local customers are unlikely to see the same cab driver twice, they may believe they will. Even if locals do not believe they will ever see a given cab driver again, they may want to encourage good drivers to stay in the market and bad drivers to leave and therefore tip to help this outcome occur. On the other hand, tipping may occur due to social norms,[1] which may vary according to the passenger's country or region of origin. If people tip due to a social norm, then tourists and locals may behave similarly. However, tourists may not know the appropriate tip amount, and as a result, they

may tip too much or too little. Given all of these mechanisms, we do not expect that locals and tourists will necessarily tip the same amount.

By examining whether there are tipping difference between locals and tourists, we can determine whether these differences in tip amounts are sufficiently large enough to cause a reallocation of taxis throughout the city. Regardless of whether the social norm effect or the enforcement mechanism is stronger, if either group tips more than the other, then this may create an incentive for taxi drivers to change where they supply their services. Therefore, we should expect an effect on the allocation of taxis throughout the city, as the drivers are likely to go where they will receive more money. This change in the allocation of taxis will affect not only where taxis are located but may also have spillover effects on other types of amenities, given that the taxi reallocation may affect the opportunity costs for the two groups. So far, most of the literature on amenities examines its importance on the attractiveness of a region for migration and location decision (Henderson 1982; Chen and Rosenthal 2008; Rodríguez-Pose and Ketterer 2012; Kuang 2017) or how important these amenities are for growth and productivity (Rappaport 2008; Rickman and Rickman 2011; Boualam 2014; Schuetz 2014). Thus, we contribute to the empirical literature by discussing the consequences in terms of the locational equilibrium of taxis due to different consumption pattern by distinct groups.

To determine the impact of differences in tipping behavior, and thus its impact on the provision of taxis throughout the city, we examine differences in the tip amounts of local residents versus tourists.[2] We draw upon a relatively novel data set of all cab rides in NYC obtained from the NYC Taxi and Limousine Commission (TLC). These data have detailed information on the cab ride, such as where the individual was picked up, dropped off, the time of pickup and drop-off, how much the ride cost, and how much the individual tipped, but we do not know anything about the attributes of the passenger. Given the lack of data on the passenger's attributes, there are likely to be unobservable differences between locals and tourists that affect tipping behavior that we cannot fully control for with our data. For example, tourists may have a higher disposable income and therefore may tip more because of an income effect. Hence, if we looked at all trips, there are likely to be unobservable variables present.

To address these concerns and to find comparable individuals, we focus on theatergoing individuals who are traveling to or from Broadway within thirty minutes of the start or finish of the show.[3] As pointed out by Seaman (2006), consumers of the performing arts tend to be more educated and have higher incomes. Therefore, by focusing on individuals who are coming to/from the Broadway district near show times, we are able to infer more about the socioeconomic attributes of the passengers and better control for unobserved attributes such as income and education. To test whether tourists tip differently than locals, we need to determine whether the passenger is a tourist. We assume that an individual is a tourist if he or she is picked up or dropped off at a hotel location.[4] In addition, since tipping may be related to the

ride and mood of the passenger, we include controls for trip distance, number of passengers, average temperature, and snow accumulation.

We find that tourists tip more than locals and that theatergoers tip more than nontheatergoers. Tourists tip .02 to .05 percentage points more than passengers who come from other locations, and theatergoers tip between .22 and .33 percentage points more than other passengers. Tourists who are also theatergoers have an aggregate tip differential of .61 to .67 percentage points from other riders. Given that the average tip is US$2.47 which represents 15.14 percent, our estimates suggest that, on average, tourists tip one cent more than nontourists, theatergoers tip four to six cents more than nontheatergoers, and tourists who are theatergoers tip eleven cents more. While these numbers are small, recall that this analysis is only for a selected portion of cab rides. Specifically, these estimates are only for rides where the rider paid with a credit card; therefore, we are not including any cash payments and tips, and we only consider rides at a specific time of day. It is possible that when all rides and all types of payment are considered, these effects are even larger.

Furthermore, while these effects may be small on a per ride basis, the benefits accumulate over time. For example, the TLC (2015, 2017) reports there were on average 426,000 trips per day and approximately 39,000 registered drivers in 2014 and 2015. Assuming all drivers split the number of trips evenly, then each driver would have around 11 trips per day, which sums to approximately 4,000 trips per year. Thus, this increase in tips represents an additional US$1.21 per day or US$440 per year for taxi drivers.

Our results are important for policy makers providing various amenities in cities. By identifying two groups—tourists and locals—we are able to determine whether these groups tip differently, which may have an impact on the allocation of taxi services in the city. For instance, Flath (2012) argues that if the amount of tipping corresponds to the Lindahl pricing of vacant cabs, then it is easier for regulators to improve the cab allocation by setting the cab fare to the marginal cost of the occupied taxis. Thus, the higher tips from tourists may change the "locus" of taxis in NYC if drivers learn and respond. This is important in terms of a locational equilibrium, and locals and/or policy makers may want to make adjustments to reach the ideal distribution of taxis.

The remainder of the article is as follows: the second section provides a conceptual model and discusses various reasons why people may tip, the third section describes the data used and the econometric model, the fourth section presents and discusses our results, and the fifth section concludes and discusses the policy implications of our research.

Conceptual Model and Motivations for Tipping

We propose the following conceptual model, based on previous models developed by Parret (2006) and Conlin, Lynn, and O'Donoghue (2003). A consumer i maximizes his or her utility (U_i) after riding in a taxi by choosing how much, in

percentage of the total amount, he or she is going to tip for that ride (T_i). We represent his or her utility function as:

$$\max_{T_i} U_i = -pT_i + \gamma_i\Big(T_i - \psi_i(t_i^0, s)|\alpha_i\Big). \tag{1}$$

Consumer *i*'s utility function captures the trade-off between paying the tip and the utility gained by tipping and conforming to the norm. The first term, pT_i, is the monetary disutility from leaving a tip, where p is the price of the ride. The second term, $\gamma_i\Big(T_i - \psi_i(t_i^0, s)|\alpha_i\Big)$, is the utility a consumer receives from tipping. The function ψ_i is the service-adjusted tip norm that depends on the initial tip belief (t_i^0) and the quality of the service (s). Note that γ_i is dependent on α_i, which indicates whether a consumer is a tourist or local. Thus, the utility in tipping comes both from tipping itself and from some psychological utility obtained from complying with a social norm. If locals tip to keep good drivers in business, the service-adjusted tip should be higher than the initial belief (assuming the default tip percentage is this initial belief).[5] In the case of tourists, the norm adherence term is dependent on the cultural background of the individual and willingness to conform to the customs of the local community. Therefore, we would not expect a priori that these two groups will derive the same utility from tipping and are likely to tip differently.

Like Parret (2006), we assume that the function γ_i is increasing and concave in $\Big(T_i - \psi_i(t_i^0, s)|\alpha_i\Big)$ and that the function ψ_i is increasing and separable. We also assume that the utility function is strictly concave. Thus, we obtain the following first- and second-order conditions from the utility maximization problem in equation (1):

$$\Lambda = \frac{\partial U_i}{\partial T_i} = -p + \gamma_i'\Big(T_i - \psi_i(t_i^0, s)|\alpha_i\Big), \tag{2}$$

and

$$\Pi = \frac{\partial \Lambda}{\partial T_i} = \gamma_i''\Big(T_i - \psi_i(t_i^0, s)|\alpha_i\Big). \tag{3}$$

Because the utility function is strictly concave in tipping, we know that $\gamma_i'' < 0$. Therefore, we know that $\Pi < 0$. Using the implicit function theorem, we can obtain the following comparative statics:

$$\partial T_i/\partial s = -\{\gamma_i''\Big(T_i - \psi_i(t_i^0, s)|\alpha_i\Big)[\partial\psi_i/\partial s|\alpha_i]/\Pi\} > 0, \tag{4}$$

$$\partial T_i/\partial t_i^0 = -\{\gamma_i''\Big(T_i - \psi_i(t_i^0, s)|\alpha_i\Big)[\partial\psi_i/\partial t_i^0|\alpha_i]/\Pi\} > 0. \tag{5}$$

Although the expected sign for each of these partial derivatives is the same for locals and tourists, we are unable to make any claim about the magnitude of these partials. Therefore, as mentioned earlier in this section, which effect is larger is theoretically ambiguous, and it is an empirical question as to which group will tip

more. Our empirical analysis will allow us to shed light on the tip difference between these groups, which should have an impact in the provision of taxi services.

Our results will have implications regarding the allocation of taxis throughout the city. If we find that locals tip more than tourists, then cab drivers are likely to respond and underprovide taxi services in the areas tourists frequent. If the under-tipping by tourists is due to not fully understanding the appropriate tip amount despite a willingness to pay, then this asymmetric information problem should prevent us from reaching the optimum allocation of cabs. Alternatively, tourists may tip more than locals. In this situation, we would see a different allocation where there is an overprovision of taxis in areas where tourists frequent. Such an outcome would suggest that locals may want to respond by adjusting tipping behavior in order to receive the optimal provision of these services. However, all of this is conditional on getting a ride. If the demand in any area is sufficiently met, then taxi drivers will have to go to another area to get business. The model described above assumes that there is excess demand in the market, which would cause some form of shifting across locations. The magnitude of these shifts, however, is likely to be partially offset by the general equilibrium effects.

Data and Econometric Model

Data

To determine whether tourists tip more or less than local consumers, we use the NYC TLC data set that contains detailed data on all taxis ride in NYC between January 2014 and July 2015. Given the extremely large number of observations in this data set, we restrict ourselves to a random 10 percent sample, containing 24,219,485 observations. The data set includes the six-digit latitude and longitude pickup and drop-off locations, miles covered, how the passenger paid, time spent in the cab, number of passengers, fare amount, tip amount, and a breakdown of costs (i.e., tolls and any Metropolitan Transportation Authority taxes). These data are available for all of 2014 and half of 2015.[6] The data set contains the population of all taxi rides during this time period.

To conduct our analysis, we must first identify which passengers are tourists and which are local consumers. As the TLC data set provides the geocoded pickup and drop-off location for each trip, we use this information to identify an individual as a tourist. Specifically, we assume that passengers who were either picked up or dropped off within 0.05 miles (approximately 264 feet) of a hotel are tourists. This cutoff is comparable to the length of an average block in NYC. As such, our cutoff reflects the notion that most tourists leaving a hotel will hail a cab within one block of the hotel. To determine where hotels are located, we used the ExcitingNY.com (http://excitingny.com/) website to obtain the location of the 363 listed hotels. To differentiate quality of the hotels, we use the Forbes classification and match the

Table 1. Day and Time of Broadway Shows.

Weekday	Starting Time	Ending Time
Tuesday	19:30/07:30 p.m.	22:00/10:00 p.m.
Wednesday	20:00/08:00 p.m.	22:00/10:30 p.m.
Thursday	19:30/07:30 p.m.	22:00/10:00 p.m.
Friday	20:00/08:00 p.m.	22:00/10:30 p.m.
Saturday	14:00/02:00 p.m.	16:00/04:00 p.m.
	20:00/08:00 p.m.	22:00/10:30 p.m.

four- and five-star hotels to our list. We assume that any individuals who were not picked up or dropped off close to a hotel are not tourists.[7]

As mentioned earlier, there are many things that affect tipping behavior, and while the TLC data set has detailed information on the ride, it does not contain information on the passenger. Given that individual attributes are likely to affect tips, we restrict our sample so that we can proxy for many socioeconomic variables. In particular, we focus on individuals who are the most likely to be consumers of the performing arts. Seaman (2006) and Lévy-Garboua and Montmarquette (2003) conduct an extensive literature review on the demand for the arts and conclude that people who consume performing arts tend to be more educated and have higher incomes. Therefore, we focus on people who are going to or from the theater district as a way of implicitly controlling for these socioeconomic attributes.

We define the theater district,[8] or Broadway, as the area within sixth and eighth avenues, between forty-first and fifty-fourth streets.[9] We focus on the performing arts because we are better able to identify theatergoers as there are clear start and end times for the shows, which is not true for art galleries or museums. We restrict our sample to cab rides that occur within thirty minutes before a show starts and after it ends, using the drop-off time and pickup time in the theater district, respectively. To do this, we use the average starting and finishing time of a Broadway show, presented in Table 1.[10] By imposing this time constraint, we restrict our sample to individuals who are the most likely to be going to Broadway for a show compared to individuals who are in the area for some other purpose.

Because shows occur all day Sundays and many shows are not performed on Mondays, we exclude Sunday and Monday cab rides. According to the Broadway schedule,[11] only six of the twenty-three shows are performed on Mondays, while from Tuesday to Saturday at least twenty of the twenty-two shows are performed each day.[12] Because the tipping data are generated for credit card payments only and do not include cash tips, we further restrict our sample to those individuals who paid by credit card. According to TLC, around 60 percent of taxi payments are made using credit cards. Because we are missing 40 percent of transactions, we are estimating a lower bound on the total differential between tourists and locals. After making all these restrictions, our sample still includes more than two million observations.

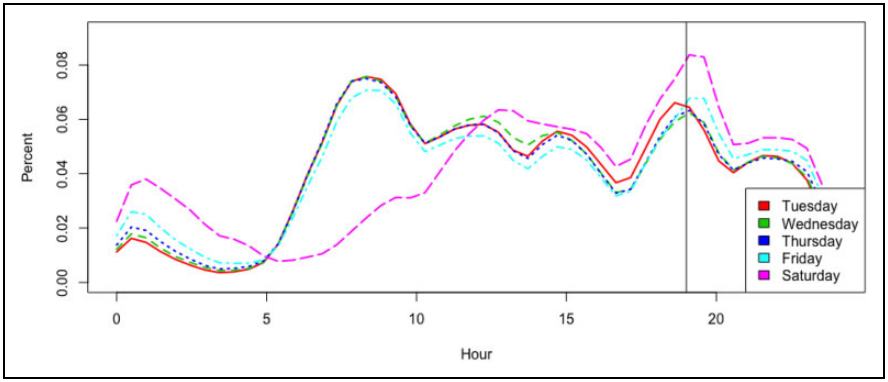

Figure 1. Trips going to theater district by fifteen-minute interval.

Figure 1 presents the distribution of rides by hour for cab rides going to Broadway on Tuesday through Saturday. From Tuesday to Friday, we observe a spike around 7 p.m. (hour nineteen), especially on Fridays which we would expect to be the evening when the most individuals attend a show. There are fewer morning commutes on Saturday, but Saturday evening still displays a spike in taxi rides around 7 p.m., when most shows begin.

Another factor that may influence tipping behavior is the mood of passenger, possibly due to weather conditions. For example, if it is raining, cab passengers may be overly appreciative because the individual was able to avoid walking in the rain and may tip more than he would on a sunny day. We obtain data for the average temperature, precipitation, and snow accumulation from National Oceanic and Atmospheric Administration and control for the effects of weather. Table 2 presents the descriptive statistics for all variables in our data set,[13] and Appendix Table A1 provides a description and the source of each variable.

Before performing our econometric analysis, it is interesting to explore our sample with respect to our outcome variable, that is, tip amount. To do so, we map the tipping patterns of locals and tourists to see whether they are similar. Figures 2 and 3 show the tip amounts for locals and tourists, respectively; and Figures 4 and 5 show the tipping percentage of the fare amount for locals and tourists, respectively. The four maps suggest that there are differences between local and tourists' tip amounts, as the hot spots vary between the groups. More specifically, locals tip the most in the Midtown area, whereas tourists tip more in the southwest corner of the island and near Times Square. The tip percentage figures (4 and 5) show a different pattern. The tip percentage hot spots for tourists are spread all over the city, possibly because hotels are spread around New York. As for locals, we notice three hot spots: one in the east of Central Park, one in Midtown, and one in the Chelsea–Gramercy area. The Midtown area is known for business, and the other two are known residential areas in the city.

Table 2. Descriptive Statistics.

Statistic	*N*	Mean	Standard Deviation	Minimum	Maximum
Number of passenger	2,120,061	1.688	1.343	1.000	6.000
Trip distance	2,120,061	2.873	2.888	0.200	19.400
Fare amount	2,120,061	12.365	8.359	3.500	52.000
Tip amount	2,120,061	2.471	1.912	0.000	17.400
Total amount	2,120,061	16.235	10.579	4.000	86.190
To theater district	2,120,061	0.068	0.251	0.000	1.000
From theater district	2,120,061	0.096	0.294	0.000	1.000
From a hotel	2,120,061	0.199	0.400	0.000	1.000
To a hotel	2,120,061	0.166	0.372	0.000	1.000
Average temperature	2,120,061	51.378	19.255	10.500	84.500
Precipitation	2,120,061	0.137	0.384	0.000	4.970
New snow	2,120,061	0.155	0.873	0.000	11.000
Snow depth	2,120,061	1.651	3.808	0.000	19.000
Last minute	2,120,061	0.003	0.057	0.000	1.000
From top hotel	2,120,061	0.016	0.124	0.000	1.000
To top hotel	2,120,061	0.011	0.105	0.000	1.000
Tip percentage	2,120,061	0.151	0.048	0.000	0.769
Tip dummy	2,120,061	0.971	0.167	0.000	1.000
From national	1,646,593	0.050	0.218	0.000	1.000
To national	1,646,593	0.041	0.197	0.000	1.000
From international	1,646,593	0.031	0.174	0.000	1.000
To international	1,646,593	0.026	0.159	0.000	1.000

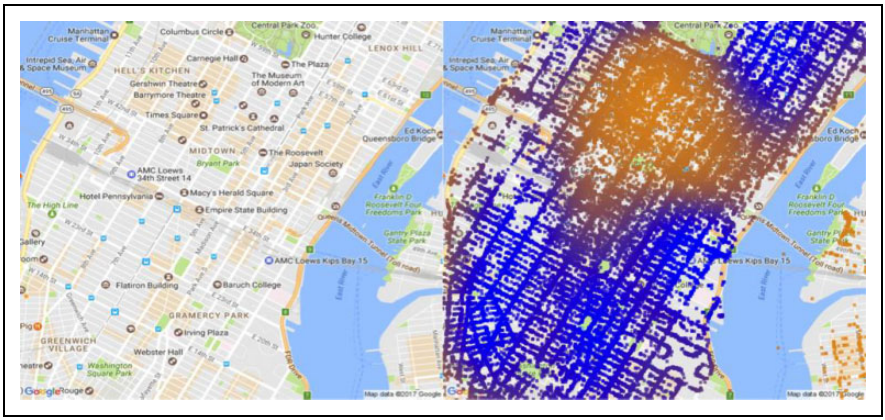

Figure 2. Tipping amount pattern of locals. The scale goes from blue to orange. Areas with higher concentration of orange mean the higher the tip amount paid.

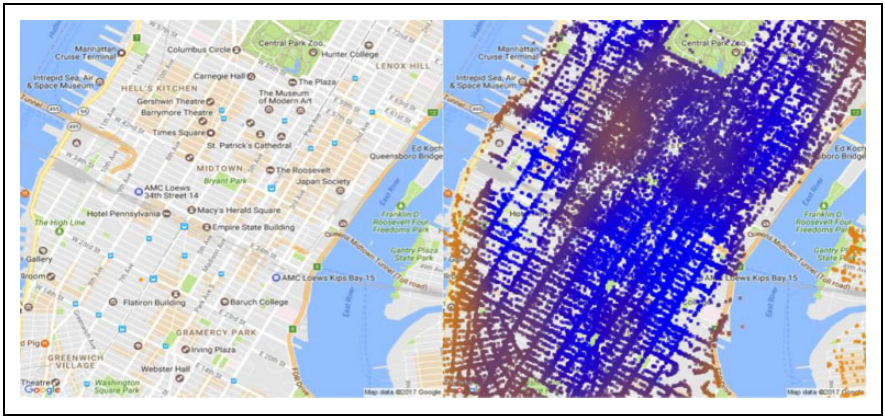

Figure 3. Tipping amount pattern of tourist. The scale goes from blue to orange. Areas with higher concentration of orange mean the higher the tip amount paid.

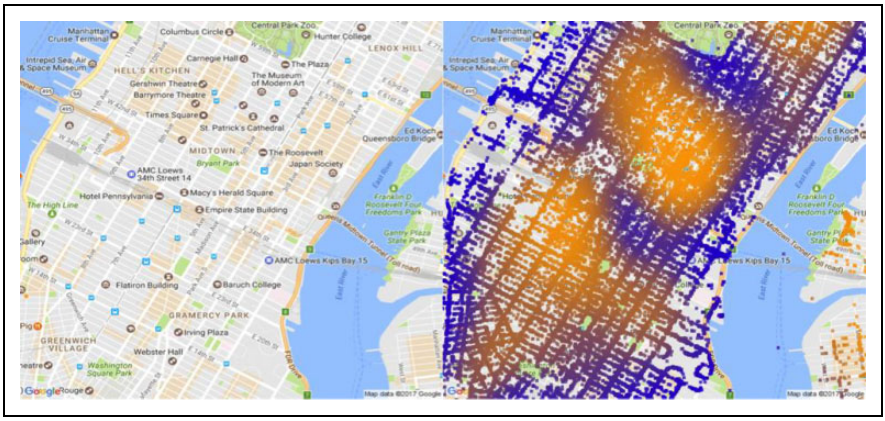

Figure 4. Tipping percentage pattern of locals. The scale goes from blue to orange. Areas with higher concentration of orange mean the higher the tip percentage paid.

Econometric Model

To estimate the effect of these characteristics on tip amount, we estimate the following model:

$$tip_i = \beta_0 + \beta_1 Tourist_i + \beta_2 Theater_i + \beta_3 (Tourist_i \times Theater_i) + X_i\delta + \varepsilon_i.$$

Where tip_i is the tip amount paid on trip i. $Tourist_i$ is an indicator variable equal to 1 if we classify the individual as a tourist. $Theater_i$ is an indicator equal to 1 if we classify the individual as a theatergoer. We also include a variety of control variables, X_i, including trip distance, passenger count, if the passenger was dropped off

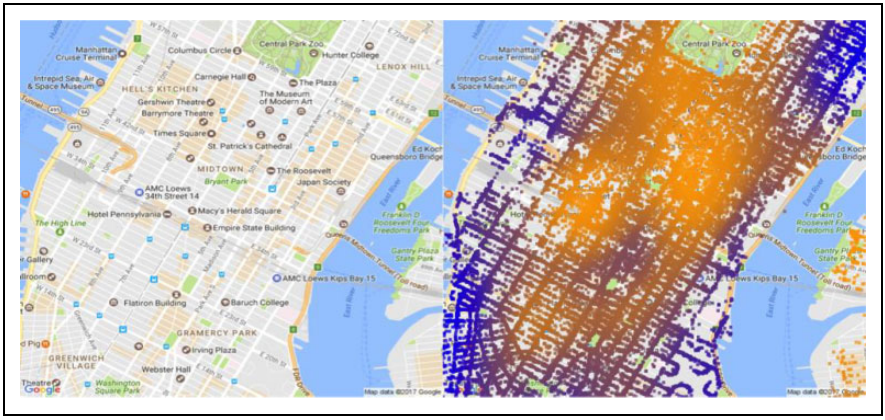

Figure 5. Tipping percentage pattern of tourist. The scale goes from blue to orange. Areas with higher concentration of orange mean the higher the tip percentage paid.

within five minutes of the start of the show, average temperature,[14] new snowfall, and snow depth. ε_i is an idiosyncratic error term.

In this model, β_1 gives the difference in tip amount for tourists relative to all other passengers, and β_2 is the difference in tip amount for theatergoers relative to other passengers. Our primary coefficient of interest, β_3, gives the relative difference in tip amount between tourists going to the theater district and all others. To further control for possible income differences of the passengers, we use the quality of the hotel as a proxy for the wealth and preferences of the customers that we classify as tourists. We include dummy variables if the passenger was coming from/going to a top hotel, that is, a hotel rated a four- or five-star hotel by Forbes.

Initially, we estimate the above model using ordinary least square (OLS). However, approximately one-third of passengers leave no tip, suggesting that the data may be truncated. To address this, we first employ a Tobit model to account for the censoring at zero as the observed zeros may be true zeros, that is, they represent individuals who choose not to tip. However, the Tobit model does not explicitly address a potential sample selection wherein individuals who do not tip may be different from individuals who do tip.[15] To address this, we use the Heckman two-stage sample selection correction; the decision to tip is modeled in the first state, and the conditional probability of tipping is used to control for potential sample selection in the second-stage estimates. This restriction allows for the zero tip to be either a true value or an unobserved value, where an unobserved value could occur for instance if the rider gives a cash tip. The Heckit model has been used many times in the previous literature to address this selection problem (see, e.g., Heckman and Sedlacek 1985; Badel and Peña 2011; Winters, Dixon, and Greene 2012; Jiménez et al. 2014).

Table 3. Ordinary Least Square Regression Results.

	Dependent Variable: Tip Percentage				
Variables	(1)	(2)	(3)	(4)	(5)
From hotel	0.089***	0.065***	0.065***	0.058***	0.051***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
To hotel	0.078***	0.050***	0.050***	0.049***	0.053***
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
To theater	0.150***	0.154***	0.147***	0.154***	0.158***
	(0.016)	(0.017)	(0.016)	(0.017)	(0.016)
From theater	0.207***	0.195***	0.194***	0.195***	0.181***
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)
From hotel × to theater	0.278***	0.200***	0.199***	0.200***	0.202***
	(0.035)	(0.035)	(0.035)	(0.035)	(0.032)
To hotel × from theater	0.281***	0.227***	0.226***	0.226***	0.230***
To top hotel	(0.031)	(0.031)	(0.031)	(0.031)	(0.030)
				0.021	0.022
				(0.033)	(0.032)
From top hotel				0.086***	0.079***
				(0.028)	(0.028)
Cab controls	No	Yes	Yes	Yes	Yes
Weather controls	No	No	Yes	Yes	Yes
Month and day fixed effect	No	No	No	No	Yes
Observations	2,120,061	2,120,061	2,120,061	2,120,061	2,120,061
R^2	.001	.003	.003	.003	.004
Adjusted R^2	.001	.003	.003	.003	.004

Note: Cab controls: trip distance, number of passengers, and last minute; weather controls: average temperature, precipitation, new snow, and snow depth. Top hotels are those with four or five stars. Robust standard errors are in parenthesis.
*$p < .1$.
**$p < .05$.
***$p < .01$.

Results

Our main results are presented in Tables 3–6.[16] In the interest of space, we only show the variables of interest: whether the individual is a tourist, going to the theater district, and indicators for the quality of the hotel.[17] First, we present our baseline OLS results in Table 3. Column 1 has no controls other than the indicators for hotel, theatergoer, and their interaction. Column 2 includes trip controls such as trip distance, number of passengers, and if the passenger was dropped off within five minutes of a show beginning. Column 3 includes the weather controls (average temperature, precipitation, new snow, and snow depth). Column 4 differentiates high-quality hotels, those with four or five stars on the Forbes website. Lastly, column 5 adds month and day of the week fixed effects.

Table 4. Tobit Results.

	Dependent Variable: Tip Percentage				
Variables	(1)	(2)	(3)	(4)	(5)
From hotel	0.093***	0.068***	0.068***	0.061***	0.054***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
To hotel	0.079***	0.050***	0.050***	0.050***	0.053***
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
To theater	0.135***	0.139***	0.132***	0.140***	0.144***
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
From theater	0.206***	0.193***	0.192***	0.194***	0.179***
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)
From hotel × to theater	0.270***	0.191***	0.190***	0.190***	0.193***
	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)
To hotel × from theater	0.274***	0.218***	0.217***	0.218***	0.222***
	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
To top hotel				0.020	0.022
				(0.033)	(0.033)
From top hotel				0.085***	0.078***
				(0.031)	(0.029)
logSigma	1.603***	1.602***	1.602***	1.602***	1.602***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Cab controls	No	Yes	Yes	Yes	Yes
Weather controls	No	No	Yes	Yes	Yes
Month and day fixed effect	No	No	No	No	Yes
Observations	2,120,061	2,120,061	2,120,061	2,120,061	2,120,061
Log likelihood	−6,326,800	−6,324,090	−6,323,941	−6,323,933	−6,323,183
Akaike information criterion	12,653,616	12,648,201	12,647,909	12,647,900	12,646,430
Bayesian information criterion	12,653,717	12,648,339	12,648,085	12,648,114	12,646,832

Note: Cab controls: trip distance, number of passengers, and last minute; weather controls: Average temperature, precipitation, new snow, and snow depth. Top hotels are those with four or five stars. Robust standard errors are in parenthesis.
*$p < .1$.
**$p < .05$.
***$p < .01$.

Looking across the five columns of Table 3, we see that the coefficients are similar across the specifications, though we have a higher R^2 with the additional controls. Given that the coefficients are similar, we focus our discussion on column 5, as this includes the most control variables and fixed effects. We see in column 5 that tourists tip approximately .05 percentage points more than nontourists and that theatergoers tip .16 to .18 percentage points more than nontheatergoers, depending if

Table 5. Heckit Results.

	Dependent Variable: Tip Percentage				
Variables	(1)	(2)	(3)	(4)	(5)
From hotel	0.053***	0.035***	0.035***	0.027***	0.024***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
To hotel	0.070***	0.049***	0.050***	0.048***	0.050***
	(0.008)	(0.008)	(0.008)	(0.009)	(0.009)
To theater	0.331***	0.328***	0.328***	0.328***	0.329***
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
From theater	0.229***	0.220***	0.219***	0.221***	0.214***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
From hotel × to theater	0.388***	0.332***	0.331***	0.331***	0.332***
	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
To hotel × from theater	0.380***	0.342***	0.341***	0.341***	0.343***
	(0.027)	(0.027)	(0.027)	(0.027)	(0.027)
To top hotel				0.025	0.025
				(0.028)	(0.029)
From top hotel				0.100***	0.095***
				(0.024)	(0.024)
Cab controls	No	Yes	Yes	Yes	Yes
Weather controls	No	No	Yes	Yes	Yes
Month and day fixed effect	No	No	No	No	Yes
Observations	2,120,061	2,120,061	2,120,061	2,120,061	2,120,061
Log likelihood	−6,122,070	−6,120,302	−6,120,112	−6,120,103	−6,119,646
ρ	−0.027	−0.026	−0.027	−0.027	−0.26
	(0.003)	(0.003)	(0.003)	(0.003)	(0.009)
	Selection Equation				
Number of passenger	−0.031***	−0.031***	−0.031***	−0.031***	−0.031***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Precipitation	−0.010**	−0.010**	−0.011**	−0.011**	−0.011**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Constant	1.958***	1.958***	1.958***	1.958***	1.958***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)

Note: Cab controls: trip distance and last minute; weather controls: average temperature, new snow, and snow depth. Top hotels are those with four or five stars. Robust standard errors are in parenthesis.
*$p < .1$.
**$p < .05$.
***$p < .01$.

they are going to or coming back from the theater. Tourists who go to the theater tip even more than nontheatergoers locals, approximately .41 to .46 percentage points. Tourists at high-end hotels tip more when they leave the hotel but not when they return.

Table 6. Falsification Results.

	Dependent Variable: Tip Percentage		
Variables	OLS	Tobit	Heckit
From hotel (t < .05)	0.051***	0.054***	0.024***
	(0.009)	(0.009)	(0.008)
To hotel (t < .05)	0.053***	0.053***	0.050***
	(0.010)	(0.010)	(0.009)
To theater	0.158***	0.144***	0.329***
	(0.016)	(0.015)	(0.014)
From theater	0.181***	0.179***	0.214***
	(0.013)	(0.013)	(0.011)
From hotel × to theater (t < .05)	0.202***	0.193***	0.332***
	(0.032)	(0.030)	(0.029)
To hotel × from theater (t < .05)	0.230***	0.222***	0.343***
	(0.030)	(0.029)	(0.027)
From hotel (t > .05 and t < .10)	−0.219	−0.241	0.019
	(0.274)	(0.268)	(0.230)
To hotel (t > .05 and t < .10)	0.589**	0.596**	0.550**
	(0.294)	(0.267)	(0.306)
From hotel × to theater (t > .05 and t < .10)	0.603	0.681	−0.264
	(1.023)	(1.068)	(1.022)
To hotel × from theater (t > .05 and t < .10)	−0.558	−0.590	−0.215
	(1.094)	(1.003)	(0.997)
Cab controls	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes
Month and day fixed effect	Yes	Yes	Yes
Observations	2,120,061	2,120,061	2,120,061
Log likelihood		−6,323,181	−6,119,644
R^2	.004		

Note: Cab controls: trip distance and last minute; weather controls: average temperature, new snow, and snow depth. Top hotels are those with four or five stars. Robust standard errors are in parenthesis. OLS = ordinary least square.
*p < .1.
**p < .05.
***p < .01.

Results using a Tobit model are presented in Table 4. We see in Table 4 that the coefficients have a similar pattern with regard to the sign of the effect and statistical significance. However, the estimated magnitude of the effect is higher than the OLS counterparts in Table 3, suggesting that the OLS estimates have a slight downward bias due to the zeros, consistent with our expectations. Additionally, the results remain statistically significant with the inclusion of month and day of week fixed effect.

As we can see from the distribution of tips shown in Figure 6, there is a significant gap in the distribution between those who tip zero and those who tip a positive

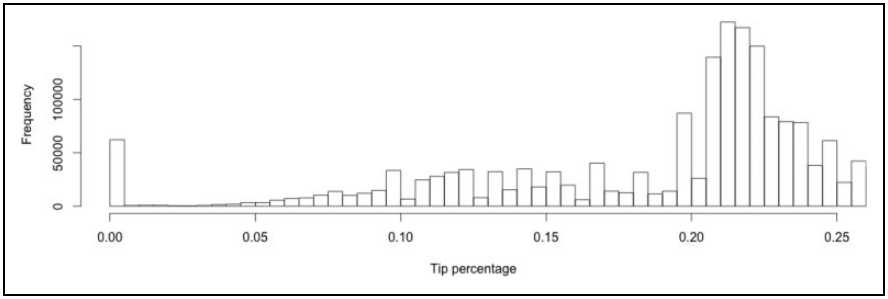

Figure 6. Histogram of tip percentage.

amount. This type of distribution suggests that it is not so much censoring at zero that is the problem, but that there is a selection process between zero and positive values. Therefore, a selection model, such as a Heckit model, is likely to be the appropriate specification. The results for the Heckman selection model are presented in Table 5.[18] As we see in Table 5, the results are similar in terms of the sign and significance levels of the coefficients to Tables 3 and 4. Tourists tip between .03 and .05 percentage points more, theatergoers tip .21 to .33 percentage points more, and tourists who are theatergoers tip .61 to .69 percentage points more than nontheatergoing locals. Tourists in four- and five-star hotels tip .10 percentage points more and again only when they are leaving the hotel.[19]

At first glance, the results above appear to be small in magnitude. However, when compared to the results of other determinants[20] of tipping, such as number of passengers (−.042), trip distance (−.060), and weather (new snow: .052; average temp: −.001), the above effects are similar in magnitude. Therefore, we can affirm that these differentials between tourists and locals are indeed important relative to other variables.

Our results that theatergoers tip more than nontheatergoers corroborate the literature suggesting education and income are related to tipping, given that theatergoers tend to be more educated and have higher incomes. For example, Azar (2005) argues that tipping is created in occupations where the consumer can show gratitude or when there is a wage differential between the consumer and worker. Therefore, as theatergoers tend to have higher education levels and higher incomes, we expect they will tip more than nontheatergoers. With regard to tourists and their tipping behavior relative to locals, our findings suggest tourists tip more than locals. This higher tip amount could be due to several factors. For example, it could be due to a higher level of gratitude, possibly because a tourist is more likely to be unfamiliar with the city, or an increased willingness to spend when on vacation. Also, it could be that tourists are overestimating what the appropriate tip amount is, and as a result, they are overtipping without realizing their error. However, for our analysis, the reasons that tourists overtip are not important.

Falsification Test

One concern with our results is that the dummy variable identifying tourists is not identifying tourists but rather anyone who happens to hail a cab near a hotel. In Table 6, we present a falsification test similar to Linden and Rockoff (2008). As previously discussed, we identified tourists as those who were picked up or dropped off within 0.05 miles of a hotel. Now, we identify a comparison group: those individuals picked up or dropped off between 0.05 and 0.1 miles of a hotel. In other words, we are comparing those individuals who are picked up or dropped off on the adjacent blocks to the hotel.

We believe that those individuals who are picked up or dropped off within 0.05 miles of a hotel are more likely to be tourists than those 0.05 to 0.1 miles from a hotel because they are closer to the hotel. However, if we are not picking up tourists, but rather locals who happen to be near hotels, then both groups should behave in a similar manner. In Table 6, we do not find a statistically significant difference between pickups and drop-offs slightly further away from the hotel from the rest of the city. Given the large sample size, the lack of statistical significance for pickups and drop-offs between 0.05 and 1 miles of a hotel is noteworthy. This suggests that there really is something unique about the individuals who are picked up within 0.05 miles of a hotel and that these individuals are systematically behaving differently than other customers. Overall, this falsification test suggests that we are capturing tourists when we classify those pickups and drop-offs within 0.05 miles of a hotel as a tourist.

Sensitivity Analysis

Because hotels are not randomly assigned, taxis could be locating in specific areas that tend to be busier due to tourists or professionals. In addition, it is known that many hotels have attendants that typically hail taxis for hotel guests. Thus, we consider smaller distances to hotels. For this sensitivity analysis, we consider tourists those individuals picked up or dropped off within 0.01, 0.02, 0.03, 0.04, and 0.05 miles of a hotel.

Table 7 presents the results for these different distances using the Heckit estimator. Regardless of the distance considered, the results remain similar in terms of magnitude and statistical significance. This reinforces the main finding that people within the block of a hotel are unique, which we infer to mean that these individuals are more likely to be tourists than locals. In addition, comparing the log likelihood for each model suggests that the baseline (0.05 miles) is the most appropriate one.

Cultural Differences in Social Norms

Because we do not have information on the passengers, it is not possible to control for differences in cultural norms, which may be correlated with tipping behavior

Table 7. Sensitivity Analysis—Heckit.

	Dependent Variable: Tip Percentage				
Variables	0.01 miles	0.02 miles	0.03 miles	0.04 miles	0.05 miles
From hotel	0.109***	0.078***	0.070***	0.041***	0.024***
	(0.029)	(0.015)	(0.011)	(0.009)	(0.008)
To hotel	0.049***	0.049***	0.049***	0.050***	0.050***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
To theater	0.401***	0.389***	0.374***	0.351***	0.329***
	(0.013)	(0.013)	(0.013)	(0.014)	(0.014)
From theater	0.223***	0.218***	0.213***	0.213***	0.214***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
From hotel × to theater	0.262**	0.298***	0.293***	0.332***	0.332***
	(0.104)	(0.055)	(0.040)	(0.033)	(0.029)
To hotel × from theater	0.355***	0.353***	0.350***	0.345***	0.343***
	(0.027)	(0.027)	(0.027)	(0.027)	(0.027)
Cab controls	Yes	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes	Yes
Month and day fixed effect	Yes	Yes	Yes	Yes	Yes
Observations	2,120,061	2,120,061	2,120,061	2,120,061	2,120,061
Log likelihood	−6,119,723	−6,119,697	−6,119,670	−6,119,653	−6,119,646

Note: Cab controls: trip distance, number of passengers, and last minute; weather controls: average temperature, precipitation, new snow, and snow depth. Top hotels are those with four or five stars. The sensitivity analysis was also done using the OLS and Tobit estimators and have similar results. These are available upon request. Robust standard errors are in parenthesis.
*$p < .1$.
**$p < .05$.
***$p < .01$.

(Greenberg 2014). Specifically, tipping tends to be an American custom, so one reason we may see differences in behavior is that there are cultural differences between the individuals we classify as locals versus tourists. To mitigate this problem, we randomly selected 100 hotels of the 363 hotels in our list and scraped hotel reviews using tripadvisor.com for the 2014 period. We use the location, more specifically city and country, in the profile of the commentator to infer whether the individual is likely to be a domestic or international guest. We then split our sample into hotels with more international guests and hotels with more domestic guests. From the 100 hotels we obtained information from, we classified 32 hotels as having primarily international patrons and 68 as primarily domestic.

Table 8 shows the results considering only the tourists for these 100 hotels and the nontourist passengers. Focusing on the Heckit model in column 3, we see that tourists coming from hotels with more international guests going to the theater district and tip more than tourists coming from hotels with more domestic guests.

Table 8. International versus National Results.

	Dependent Variable: Tip Percentage		
Variables	OLS	Tobit	Heckit
To theater	0.138***	0.124***	0.291***
	(0.018)	(0.018)	(0.016)
From theater	0.180***	0.179***	0.199***
	(0.015)	(0.015)	(0.013)
From national	−0.011	−0.009	−0.036**
	(0.019)	(0.020)	(0.016)
To national	0.102***	0.103***	0.092***
	(0.021)	(0.021)	(0.018)
From national × to theater	0.287***	0.278***	0.417***
	(0.069)	(0.065)	(0.061)
To national × from theater	0.170**	0.160**	0.312***
	(0.066)	(0.064)	(0.058)
From international	0.066***	0.068***	0.043**
	(0.023)	(0.024)	(0.019)
To international	−0.066***	−0.066**	−0.068***
	(0.025)	(0.026)	(0.022)
From international × to theater	0.361***	0.355***	0.461***
	(0.091)	(0.086)	(0.080)
To international × from theater	0.240***	0.236***	0.302***
	(0.086)	(0.083)	(0.080)
Cab controls	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes
Month and day fixed effect	Yes	Yes	Yes
Observations	1,646,593	1,646,593	1,646,593
Log likelihood		−4,910,178	−4,752,479
R^2	.004		

Note: cab controls: trip distance and last minute; weather controls: average temperature, new snow, and snow depth. Top hotels are those with four or five stars. Robust standard errors are in parenthesis. OLS = ordinary least square.
*$p < .1$.
**$p < .05$.
***$p < .01$.

When returning to the hotel, tourists returning to more domestic hotels tip more than tourists returning to more international hotels. Overall, we do not find systematic evidence that international customers are tipping less, suggesting that tourists are learning trying to adapt to social norms.[21]

Conclusions and Policy Implications

In this article, we examined whether there are differences in tipping behavior between locals and tourists. To accomplish this, we focus on NYC and use a novel

data set from the TLC. Using geo-coordinates of the pickup and drop-off locations, we identify which passengers are more likely to be tourists. Our analysis suggests that tourists tip more than locals and that theatergoers tip more than nontheatergoers. Specifically, the results show that tourists tip .03 to .05 percentage points more, theatergoers tip .21 to .33 percentage points more, and tourists who are theatergoers tip between .61 and .69 percentage points more. In monetary terms, tourists who are theatergoers tip eleven cents more than locals, where the average tip is US$2.47 dollars. These results can be considered lower bound estimates, since our results rely only on credit card payments.[22] Moreover, our findings are consistent across several specifications, in which we control for possible sample selection and censoring.

Overall, we find that tourists are tipping more than locals. This finding has important implications for the allocation of taxis throughout the city. Specifically, because tourists tip more than locals, cab drivers are likely to adjust and provide these services more in areas with tourists than in areas with locals, resulting in what may be a suboptimal allocation of taxis. Policy makers and local residents should be interested in this finding, as it suggests that some adjustment is needed to reach a more desirable allocation of cabs for local residents. Although our results do not allow us to confirm this hypothesis, the overtipping may act as an extra incentive to the supply of taxis, as drivers earnings from tipping would also be higher.

We can speculate as to what our results suggest regarding the provision of other endogenous amenities in the city. Assuming the provision of cabs is not affected but there is a change in the distribution of taxis in the city, one could expect that the overall quantity of endogenous amenities should remain similar, as there is no change in costs to both suppliers and consumers. However, if there is a shift in cab allocation to areas where there are more tourists, suppliers will have the incentive to provide more services in this area for this group due to the lower access costs.

There are three main policy implications of our analysis. First, because tourists tip taxi drivers more than locals, the impact of tourists on a local economy may be underestimated, as tips are frequently not reported income and therefore are not included when calculating multiplier effects of tourism on the economy. However, this should not be extrapolated to other types of tipping behavior, such as in restaurants, as those individuals who take taxis are likely to be higher income on average. Second, we find evidence that wealthier and more educated people are likely to tip more because we find theatergoers tip more than nontheatergoers, and previous research has found theatergoers tend to have higher incomes and education levels. This suggests that the provision of cultural amenities similar to those found on Broadway should be impacted as well and increase due to lower opportunity cost in transportation. Third, and most importantly, if the extra tip is sufficient to change the behavior of taxi drivers, these individuals might change their locus of work. This suggests that locals need to adapt to correct this change by taxis or that policy makers may need to get involved to help correct this inefficiency.

Future research should focus on obtaining more information from both passengers and drivers, as well as information on cash tipping. One of the caveats of our

analysis is that we only observe credit card tips. Therefore, the difference in tipping behavior we have observed may change once we account for those who pay or tip in cash. With better information on passengers and drivers, we would be able to tease out the determinants of tipping in a more comprehensive way. Also, information on cash tips would provide additional information on behavior, since then we could control for the payment method which the previous literature (Lynn 2006) has found influences tipping. As shown by Shefrin and Nicols (2014), demographics and socioeconomic status may influence cash and credit card use; however, as previously explained, we focus on theatergoers to control for socioeconomic status as they are usually more educated and have higher income. Another caveat of our analysis is the rise of ride-sharing services such as Uber and Lyft that can affect the use of taxis. Future work should tie in data on these services as it becomes available, as this will further show how these other options have affected behavior in the taxi market.

Appendix

Table A1. Description and Source of Variables.

Variable	Description	Source
Number of passenger	Number of passengers in trip	TLC
Trip distance	Total trip distance	TLC
Pickup longitude	Pickup longitude for trip	TLC
Pickup latitude	Pickup latitude for trip	TLC
Drop-off longitude	Drop-off longitude for trip	TLC
Drop-off latitude	Drop-off latitude for trip	TLC
Fare amount	Trip fare amount	TLC
Tip amount	Tip amount paid	TLC
Total amount	Total amount paid	TLC
To theater district	Binary variable, equals 1 if drop-off location is theater district	TLC/Broadway
From theater district	Binary variable, equals 1 if pickup location is theater district	TLC/Broadway
From a hotel	Binary variable, equals 1 if pickup location is 0.05 miles from hotel	TLC/Hotel
To a hotel	Binary variable, equals 1 if drop-off location is 0.05 miles from hotel	TLC/Hotel
From top hotel	Binary variable, equals 1 if pickup location is four or five star hotel	TLC/Forbes
To top hotel	Binary variable, equals 1 if drop-off location is four or five star hotel	TLC/Forbes
Maximum temperature	Daily maximum temperature in Fahrenheit	NOAA

(continued)

Table A1. *(continued)*

Variable	Description	Source
Number of passenger	Number of passengers in trip	TLC
Minimum temperature	Daily minimum temperature in Fahrenheit	NOAA
Average temperature	Daily average temperature in Fahrenheit	NOAA
Precipitation	Daily precipitation in inches	NOAA
New snow	Daily snowfall in inches	NOAA
Snow depth	Daily snow depth in inches	NOAA
Last minute	Binary variable, equals 1 if drop-off time is five minutes before or after show	TLC/Broadway
Tip percentage	Percentage of tip from total amount paid	TLC
Tip dummy	Binary variable, equals 1 if there is a tip	TLC

Note: NYC TLC = New York City Taxi and Limousine Commission; Broadway = www.broadway.com; Hotel = http://excitingny.com/ny-hotel-list.shtml; Forbes = Forbes hotel classification; NOAA = National Oceanic and Atmospheric Administration.

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Notes

1. While in the United States this social norm argument is likely to be present and important, in other countries tipping is not as common of a practice. As a robustness check, we attempt to separate American versus international tourists to address concerns that there may be cultural differences driving tipping behavior.
2. The objective of this article is not to separate and fully differentiate the possible mechanisms behind tipping by tourists and locals. For a more detail discussion on why people tip, see Ben-Zion and Karni (1977), Azar (2003), Azar (2004a, 2004b), Azar (2007), and Greenberg (2014).
3. We define Broadway as the area between sixth and eighth avenues and between forty-first and fifty-fourth streets.
4. Tourists are those who have their ride starting or finishing 0.05 miles from a hotel, which were obtained at (http://excitingny.com/) website.
5. For instance, Haggag and Paci (2014) provide evidence that the default suggestions given in some taxis (i.e., 20 percent, 25 percent, or 30 percent) when paying with credit cards have a positive impact on the amount tipped.

6. These data can be downloaded at http://www.nyc.gov/
7. We acknowledge that tourists may take cabs to get from one site to another that is not a hotel or the theater district. However, we do not believe this will bias our results due to of the time of the day restriction we make (roughly, between 7 p.m. and 8 p.m. and between 9:30 p.m. and 10:30 p.m.). During this time, it is more likely that tourists are leaving the hotel to go to a show or returning to the hotel from a show. If we were using every hour of the day, this would more of a concern.
8. See, http://www.nyc.com/visitor_guide/theater_district.702514/editorial_review.aspx, for more details.
9. While this area does have some financial, fashion, and commercial industries, given our time of day restrictions, these places are likely closed and therefore there are unlikely to be a significant number of people traveling to and from these places. However, there are also numerous bars and restaurants in this area, which people are likely to be traveling to and from at this time which could affect our results. This is true about Manhattan in general, so while people may be traveling to restaurants in this area, we do not believe that there are more going to this specific area versus others, so we do not believe that this will bias our results.
10. We obtain the list of shows and their schedule from the Broadway website.
11. The current Broadway show times can be retrieved at: http://www.playbill.com/article/weekly-schedule-of-current-broadway-shows-com-142774
12. On Tuesday one show is dark, and Wednesday two shows are dark. Sunday also has some shows that are dark, but this day is already excluded, given that shows run throughout the day.
13. Appendix Table A1 provides a full description and the source of each of our variables.
14. There is some disagreement in the literature about the influence of weather on the tipping behavior. While Cunningham (1979) and Devaraj and Patel (2017) find supporting evidence of such, Flynn and Greenberg (2012) do not.
15. For a more detailed discussion on the difference between the models, see Humphreys (2015).
16. Because we restrict our sample to rides just before and after a show, we are comparing similar rides in terms of time of day and do not need to include time fixed effects. We also do not include day of week fixed effects, because as shown in Figure 1, although there may be some difference in levels, at the time we consider every day of the week has a similar trend. Nevertheless, we ran the ordinary least square, Tobit, and Heckit models including both time and day of week fixed effects, and the results are similar.
17. Full results are available from the authors upon request.
18. As an additional robustness check, we used a double hurdle model. In this model, the observed zeros are true corner solutions. The results show the signs remain the same, and they are still statistically significant. The results are available upon request.
19. We have also estimated probit and logit models and the results follow the same pattern. These results are not shown in the interest of space but are available from the authors upon request.

20. The results in parenthesis are for the Heckman model for trip distance and weather and the Tobit model for the number of passengers.
21. The log-likelihood ratio test comparing the specification with the split sample of hotels between domestic and international hotels versus a specification in which we do not make such split shows that the former is a preferred specification. This suggests our two hotels sample are different, and we gain extra and useful information by splitting the sample.
22. Lynn and McCall (2016) argue that for restaurant, credit card tipping is usually higher than cash tipping. However, this is still an open area of research when discussing cab tipping.

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